DRONACHARYA
College of Engineering

PRACTICAL FILE

Department of CSE(AI&ML)

Operating System Lab

NAME:

BRANCH:

SEM:

ROLL NO:

Practical-1

Aim:- To study about Windows operating Systems

Theory:-

Windows Operating systems is one of the most popular operating systems and it is used by millions of people all over the world. This os was developed by "Microsoft Corporation" in 1985 and they take more than 90% of the market share from the global market in the IT sector (Computer operating System). The reason behind the popularity of this os is that it comes up with graphical user interface and with preloaded applications which makes the user easy to use and make their experience good. They can easily use any third-party application in this OS without writing any codes and do many types of things

Features in Windows Operating System

1) Desktop

2) Taskbar

3) Task Manager

4) Control Panel

5) Command Prompt

6) Setting

7) Cortona

8) Start Menu

1) Desktop:

Desktop is a by default GUI based screen provided provided by the Microsoft in windows OS. It is a place where all the software applications icons icon are available on a screen, so the were can do their work by clicking on these applications.

2) Taskbar:

Taskbar is an area where you can check which application in your PC/Desktop is currently open er which is met. given image, you can lee you can lee that the currently open. met, like in the only chrome application is currently open

3) Task Manager:

Task Manager is a system monitoring program where you can monitor your system what's currently running, on your computer It shows your system rescue information, what number of resources can be used by each application

4) Control Panel:

It is a management tool where you can manage all the resources on your like you can all the resources on This our management tool where you can make some changes on Hardware resources (Printer, Mouse, Keyboard, etc.), networks, programs, personalization etc.

5) Command Prompt:

Command Prompt is a terminal where you can operate your computer by writing some commands like cd, cls, dir, exit, find, hostname, shutdown etc. and many others. It is mostly known as "cmd".

6) Setting:

The word "setting" you better know, this is available in all the devices smartphones, tablets, PC, etc. setting allows us to make different types of wallpaper, pictures to background, we can adjust the icon size and of your system change the time and date of your system

7) Cortona:

Cortona is an assistant which helps us to execute our program by hearing our voice, it saves our time and focuses attention on what matters most

8) Start Menu:

Start Menu shows all the applications and programs of our system It also shows the history of what application was open currently

History of Windows operating system

1983 was the year where the first window operating system was developed which is done by multi-technology software company "Microsoft" and it was founded by two popular electronics magazines friends "Bill Gates" and "Paul Allen" in 1975. After some years Paul Allen left the Microsoft company (1983) and right now the Bill Gates were the only person who owns Microsoft and there are many shareholders who invest in Microsoft Versions of windows operating system:

a) Windows 1

It was the first windows from Microsoft which is released on 20 November 1985. The was the first step from Microsoft to produce a GUI based operating system. After that they sold it in the market as $100.

b) Windows 2

After two years, Microsoft released the second version of the window which is released on 9 December 1987. It was sold for $100.

c) Windows 386

The release the date of window 2, "Window 386", was released on the same day (on 9 Dec 1987), and it, it was sold at $100.

d) Windows 3

It was released on 22 May 1990 and this was the first version of windows that required an "HDD (Hard Drive Desk) It was initially sold at a price of $149.95 and the upgraded version was sold for $79.95

e) Windows 3.1

It was released on 6 April 1992. It was most loved. version in the market, Within 2 months the millions of copies of these version were sold in the market

f) Windows 95

Windows 95 was released on 24 August 1995 This was the time where the windows craze started, within a few days the millions of copies were sold

g) Windows 98

Windows 98 was released in 1998. This was the updated version of windows 95 where the Microsoft added some new feature which was not available in windows 95.

h) Windows 2000

Windows 2000 was launched on 17 Feb 2000. These version was based on a business-oriented system

i) Windows XP

Windows XP was released on 25 October 2000 and this was the best version of Windows OS because it is a stable OS which is built on windows 2000 kernel

J) Windows Vista

In January 2006, this version was released and it is Version of Windows XP where the Vista added some additional features with its looks and feel

k) Windows 7

Windows 7 is a successor to window vista which was released on 22 July 2009. This version of windows aims to those bugs which war faced in windows Vista and make it user- Friendly OS

l) Windows 8

Window 8 was released on 26 October 2012 and this version of windows comes with some new features like faster booting speed, web store, improved search function, USB 3.0 supported device etc. and many others

m) Windows 8-1

Windows 8.1 was released on 23 august 2013 and it is little upgrade of window 8

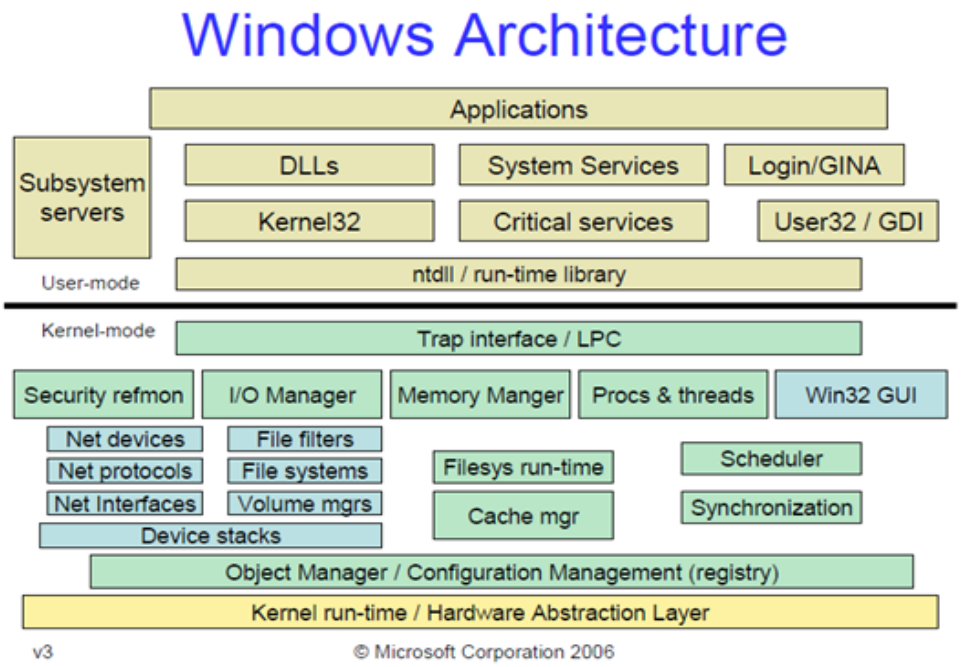
n) Windows 10

Windows 10 was released on 29 July 2015. This version comes up with some improvements and with new features.

o) Windows 11

Well, we all know window II is recently released by Microsoft It is released on 5 Oct 2021. It is also coming up with some Improvements and with some features like Apple mac-like interface, Integrated android apps, splitting the screen, Better visual desktop support and many other things

Windows OS Structure:



Practical-2

Aim:- To study about UNIX Operating System

Theory :-

UNIX is a powerful Operating System initially developed by Ken Thompson at AT&T Bell Laboratories in 1970 prevented among Scientific, Engineering and Academic institutions due to its most appropriate features discussed further, In UNIX file system is a hierarchical structure of files and directories when access can store and relative information using the files.

Features of UNIX Operating System :-

1)Multi-tasking :-

A UNIX Operating system as a multitasking Operating System that allows you to Initiate more than one so that one task is performed as foreground and other task as background progress.

2)Multi-user :-

UNIX Operating system supports more than one user to access computer resources like main memory, hard disk, tape drives etc. Multiple users can log on to the system from different terminals and run different jobs that share the resources of a command terminal

3)Portability:

This features makes the UNIX work on different machines and platforms with the easy transfer of code to any computer System. Since a significant portion of UNIX is written in C language and only a tiny portion is cooled un assembly language for specific hardware C

4)File Security and Protection:

Being a multi-user system, UNIX makes special consideration for file and security system. UNIX has different levels of Security using assigning username and password to individual users ensuring the authentication, at the level providing file access permission

5) Command Structure:

UNIX commands are easy to understand and simple to use Example - "cp", "mu' etc. while working in the UNIX environment, the UNIX commands are case and are entered un lower case

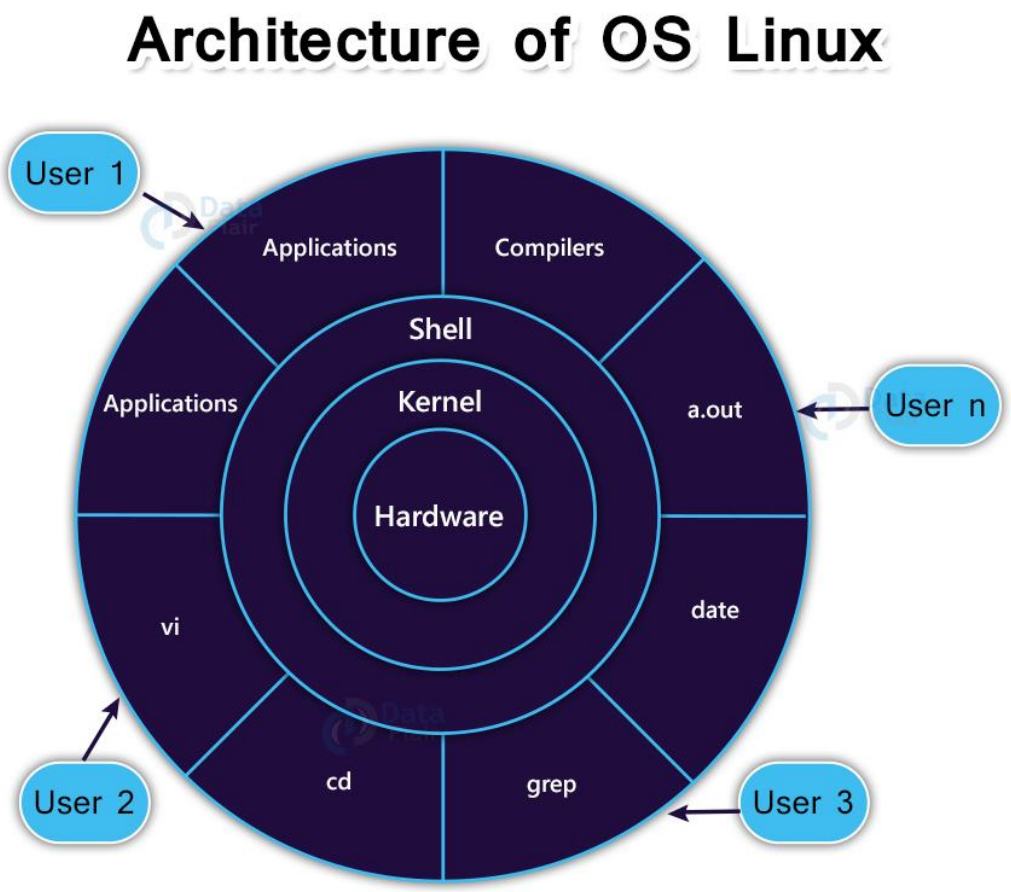
6) Communication:

In UNIX, communication is an excellent feature that enables the user to communicate worldwide It supports various communication facilities provided using the write command, mail command, talk command etc.

7) open-source:

ONIX operating system is open source it means it is freely available to all and it a community-based development project

The structure of UNIX OS Layers are as follows:



1) Layer-1: Hardware -

This layer of UNIX consists of all hardware-related information in the UNIX environment.

2) Layer-2: Kernel

The core of the operating system that's liable for maintaining full functionality is named the kernel. The kernel of UNIX runs on the particular machine hardware and interacts with the hardware effectively

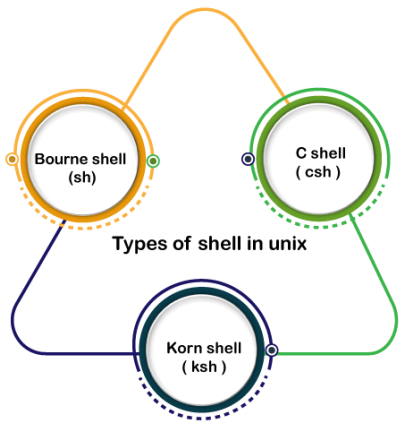
It also works as a device manager and performs valuable functions. for the processes which require access to the peripheral devices connected to the computer The kernel controls these devices through device drivers

The kernel also manages the memory Processes that have owner's humans or are executed programs systems who initiate their execution. To make effective use of main memory and to allocate a sufficient amount of memory to every process It uses essential techniques like paging, swapping and virtual storage

3) Layer-3: Shell-

The shell is an interpreter that interprets the command submitted by the user at the terminal, and calls the program you simply want a command you use the cursor keys to It also keeps a history of the list of commands you have typed in If you need to repeat you typed, use scrolls up and down the list or type history for a list of previous commands. There are various commands like cat, mu, cat, grep, id, we and many more

Types of shell in UNIX System :



a) Bourne Shell -

This shell is simply called the shell It was the first shell for It is still the most widely used shell on Unix system UNIX OS

b) C Shell-

The C shelf is another popular shell commonly available on a UNTY system. The C shell was developed by University of California at Berkely and removed some of the Shortcomings of the Baume

c)Korn Shell-

This shell was created by David Korn to address Bourne shell's user-intersection issues and to deal with shortcomings of the I shell's scripting quirks

4) Layer-4 Application Programs Layer-

It is the outermost layer that executes the given ex UNIX distributions typically come with several useful application programs.

Example - emacs editor, star office, xv image viewer, ger compiler etc.

Practical-3

Aim:- File and Directory Related commands in UNIX

1) mkdir (Make directory):-

To create a new directory

Syntax:

$ mkdir directory_name

$ mkdir dir1 dir 2 dir3

```
$  mkdir os
$  mkdir Rishi aiml  oslab
$  ls

 aiml Rishi os oslab
```

2) ls(list directory and files):-

To show the list of directories and files

Syntax:-

$ls[option]

options:-[-L,-c,-d,-r,-s,-a]

```
$ls-l
total 52
-rw-rw-r--1  dce202sys4   dce202sys4   26352  March 5   19:58  aiml
-rw-rw-r--1  dce202sys4   dce202sys4   26352  March 5   19:58  Rishi
-rw-rw-r--1  dce202sys4   dce202sys4   26352  March 5   19:58  os
-rw-rw-r--1  dce202sys4   dce202sys4   26352  March 5   19:58  oslab
$ls-c
aiml Rishi os oslab
$ls-r
oslab os Rishi aiml
```

3) cd (change directory) :-

To change the directory and path

Syntax :-

$~cd directory_name

```
$cd oslab
/oslab$
```

4) cd ~ :-

To go to home directory

Syntax :-

$cd~

```
/oslab$ cd~
$
```

5) cat(concatenation) :-

It is used to create and shows the files and content

Syntax:-

$cat>file_name(contents/data) ctrl+d

$cat file_name

```
/oslab$ cat>hello.txt
Hello everyone
I am Rishi
/oslab$ls
Hello.txt
/oslab$ cat hello.txt
Hello everyone
I am Rishi
```

6) cp (copy) :-

It is used to copy contents of file

Syntax :-

$cp[option] source_file destination file

```
/oslab$ cp s.txt.d.txt
/oslab$ cat d.txt
Hey! We are working with UNIX commands It is a cp command used to
copy contents in a file
Thank you!!
```

7) mv (move) :-

To move the contents

Syntax :-

$mv first_file_name second_file_name

```
/oslab $ mv hello.txt hi.txt
/oslab $ ls
d.txt hi.txt s.txt
/oslab $ cat hi.txt
/oslab $ cat hi.txt
Hello everyone
I am Rishi
```

8) od:-

To convert the content of input file into decimal, octal format.

Syntax :-

$ od file_name

```
/oslab $ cat > world.txt
Hello
/oslab $ od world.txt
0000000  062550  066154  005157
```

9) wc(word count):-

Display the lines, word and characters

Syntax:-

$ wc file_name

Options:- [-l,-w,-c]

```
/oslab$ wc hello.txt
2 6  35  hello.txt
/oslab $ wc-l hello.txt
2 hello.txt
/oslab$ wc-2 hello.txt
6 hello.txt
/oslab$wc-c hello.txt
35 hello.txt
```

10) grep:-

Prints lines matching a pattern

Syntax:-

$ grep word_to_be_find file_name

```
/oslab$ grep am hello.txt
I am Rishi
```

11) rm(remove file):-

To delete a file

Syntax:-

$ rm[option] file_name

```
/oslab $ rm s.txt
/oslab $ ls
d.txt hi.txt world.txt
```

$rm*:-

Removes all files in the working directory. If it write protected, you will be prompted before rm removes it

12) rmdir(remove directory):-

To delete a directory

Syntax:-

$ rmdir [option] dir_name

options:-[-I,-r,-R]

```
$ rmdir Rishi
$ ls
aiml os oslab
```

$rmdir -p mydir/mydir1 :-

This will first remove the child directory and then remove the parent directory

Practical-4

Aim: Essential Unix commands for working in UNIX environment

1) Date command:

It is used to display current date and time

Syntax:-

$ date

options: [-m-d-y.-H-D]

$ date +%m/%y/%d/%H/%D

```
$date
Saturday 5 April 2025 22:22:56 PM ist
$date+%m
02
$date+%d
15
$ date+%y
24
$date+%H
13
$date +%D
02/15/24
```

2) cal command (calender):-

It is used to display a simple calendar

Syntax :-

$ cal

```
$ cal
 April 2024

SU   Mo  Tu  We  Th  Fr   Sa
            1    2    3    4    5
6     7    8    9    10   11  12
13   14   15  16   17   18   19
20   21   22  23  24    25  26
27   28   29  30
```

3) who:-

It is used to deeplay current logging user's details

Syntax:-

$ who

```
$ who

dce 202 sys 4:0  2025-03-14 12:36 (:0)
```

4) pwd (print working directory / current directory):-

It is used to show current/working directory

Syntax:

$pwd

```
$ pwd

/ home /dce 202 sys4
```

```
/oslab $ cat file 1.txt

apple

banana
```

```
cherry
fig
grape
litchi
watermelon

/oslab $ cat > file2.txt
apple
cherry
fig
grape
litchi
```

5) head :-

It is used to display starting content

Syntax :-

$ head -5 file_name

```
/oslab $ head-2 file 1.txt
apple
banana
```

6) tail:-

It is used to display ending content

Syntax:-

$tail -5 file_name

```
/oslab $tail-3 file 2-txt
fig
grape
litchi
```

7) more and less :-

It is used to display next and previous page

Syntax:-

$ more file name

$ less file-name

```
/oslab $ more file 1. txt
apple
banana
cherry
grape
watermelon
/oslab $ less file1.txt
apple
banana
cherry
fig
grape
watermelon
file1.txt (END)
```

8) cmp (compare):-

Syntax:-

$ cmp F-file.name S_file_name

```
/oslab $ cmp file1-txt file 2-txt
file1.txt file 2-txt differ: byte 7, line 2
```

9) comm (common):-

Syntax:-

$comm F_file_name S_file_name

```
/oslab & comm file1.txt file2.txt
            apple
banana
           cherry
           Fig
          grape
          litchi
watermelon
```

10) diff (difference):-

Syntax:-

$ diff 1. F_file_name S.file_name

```
/oslab $ diff file1.txt file2.txt
2d1
< banana
7d 5
< Watermelon
```

11) w:-

It displays information about the users currently on the machine, and their processes.

Syntax:- w [options] user

Options:-

•w-h: This option don't print the header.

w-u: This option will ignore the username while figuring out the current process and CPU times

•w-s: This option uses the shoe format. It will not print the login time, JCPD and PCPU times

•W-i: This option display IP address instead of hostname from field

• W user: This option will show information about Specified user only.

Practical-5

Aim:- I/O Redirection and Piping Introduction

I/O redirection:-

I/O redirection allows you to enter the input source of a command as well as its output and error messages are sent to. And this is made possible by the “<” and ">" redirection operators

Input and Output in the Unix Environment is distributed across three streams. These streams are:

- standard input (stdin)
- standard output (stdout)
- standard error (stderr)

To see standard input in action, sun the cat program

Commands with a single bracket overwrite the destination's existing contents.

Overwrite

> - standard output

< - standard Input

2> - standard error

Commands with a double bracket do not overwrite the destination's existing contents

Append

>> - standard output

<< - standard Input

2>> - standard error

Output redirection command

command> output-file-spec

$ ls > my-files [Enter]

Notice there is no output appearing after the command, only the return of the prompt. This is because all output from this command was redirected to the file my_file.

```
/oslab $ echo “Hello world”>
file1.txt
/oslab $ cat file1.txt
Hello World
```

Input redirection command

Command<input_file_spec

```
/oslab $ wc file1.txt
    1   2  12  file1.txt

/oslab $ wc<file.txt
1   2   12
```

We can use wc in conjunction with input redirection as above.

Piping:-

Piping takes a command’soutput and transport it (pipes it) to the input of another command :-

Command 1 | command 2

Pipe command 1’s output to 2’s input

```
/oslab $ grep apple file.txt/more
Apple
/oslab $ ls-l/more
total 24
-rw-rw-r--1  dce202sys4   dce202sys4  99  March 5   19:58  d.txt
-rw-rw-r--1  dce202sys4   dce202sys4  12  March 5   19:58  file1.txt
-rw-rw-r--1  dce202sys4   dce202sys4  30  March 5   19:58  file2.txt
-rw-rw-r--1  dce202sys4   dce202sys4  32  March 5   19:58  hi.txt
-rw-rw-r--1  dce202sys4   dce202sys4   0   March 5   19:58  world.txt
```

Practical-6

Aim: To study about Vi Editor

Starting the VI editor :-

The vi editor lets a user create new files or edit existing files. The command to start the Vi editor is vi, followed by the filename. For example, to edit a file called example 1, you would type vi example 1 and then hit return key. You can start VI without a filename, but when you want to save your work, you will have to tell VI which filename to save it into later when you start VI for the first lime, you will see a screen filled with tildes (A tilde looks like this: ~) on the left side of the screen. If you started VI without a filename, the bottom line of the screen will just be blank when VI starts

To create or edit a file: vi [filename]

$ vi test

```
/oslab $ vi file2.txt
apple
cherry Fig
grape
Litchi
~
~
~
~
~
~
"file2.txt" 5L,30B
```

Modes of operation in vi editor

There are three modes of operation in vi:

- **Command Mode:** When vi starts up, it is in command mode. This mode is where vi interprets any character we type as commands

and thus does not display them in the window. This mode allows us to move through a file and to delete, copy or paste a piece of text

To enter into command mode from any other mode, It requires pressing the [esc] key. If me press [esc] when me are in command mode, then vi will keep or flash the screen.

• **Insert mode:** This mode enables you to insert text into the file.

Everything that's typed in this mode is interpreted as input and finally, it is put in the file. The vi always starts in command mode. To enter text, you must be in insert mode. To come in insert mode simply type i. To get out of insert mode, press the Esc key, which will put you back into command mode.

• **Last line mode (escape mode):** Line mode is invoked by typing a

colon (:), while vi is in command mode. The cursor will jump to to last line of the screen and vi null wait for a command. This mode enables you to perform tasks such as saving files, executing commands

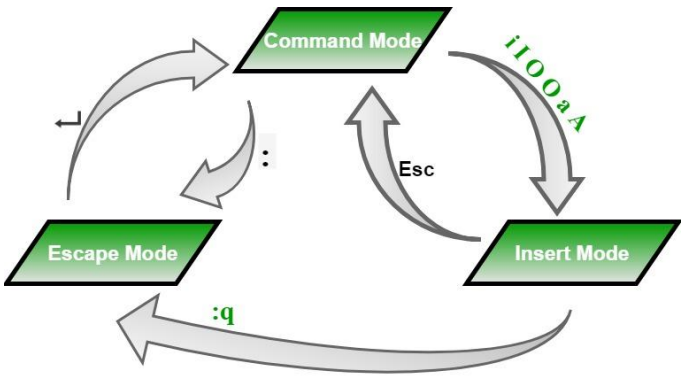
Starling the vi editor

• vi filename:

creates a new file if it already not exists, otherwise opens existing file.

• vi-R filename:

opens an existing file in read only mode.



Editing and inserting in Files (Entering and Replacing Text):

To edit the file, we need to be in the insert mode. There many ways to enter insert mode from the command mode.

- **i:** Insert texts before current cursor location
- **I:** Insert text at beginning of current line
- **a:** Insert texts after current cursor location.
- **A:** Insert text at end of current Line
- **o:** Creates a new line for text entry below cursor location
- **O:** creates a new line for text entry above cursor location
- **r:** Replaces text from the cursor to right
- **R:** Replaces single character under the cursor with the next character typed.
- **s:** Replaces single character sender the cursor with any number. of characters
- **S:** Replaces entire line.

Deleting Characters:

Here is the list of important commands which can be used to delete characters and lines in a opened file.

- **x (Lowercase):** Deletes the character at the cursor location.
- **d^:** Deletes from current cursor position to the beginning of the line.
- **d$:** Deletes from current position to the end of the line
- **Dd:** Deletes the line curser is on

Copy and Past Commands:

Copy lines or words from one place and paste them to another place by using following commands

- **Vy:** Copies the current line.

• **9yy:** Yank current line and 9 lines belon

•**p:** Puts the copied text after the cursor

•**P:** Puts the yanked text before the cursor

Save and Exit commands of the ex Mode:

Need to press [Esc] key followed by the colon (:) before typing the following commands:

•**q:** Quit

•**wq:** Write and quit (save and exlt)

•**w filename:** Write to file called fileName (rave as)

•**w! filename:** overwrite to file called fileName (save as forcefully).

Searching and Replacing in (ex Mode):

vi also has powerful search and replace capabilities The formal syntax for searching is :

: s/string.

Syntax for replacing one string with another string in the current line is:

:s/pattern/replace/

Syntax for replacing every occurence of a string in the entire text is similar. The only difference is the addition of a "60%" un front of the "s":

:%s/pattern/replace/

Read and write from a file and to a file.

:r file nome

:r !date

It will append the current date at cursor position into the file opened.

: w>> file.name

It will write the content to the given file.

Moving within a File (Navigation):

To move around within a file without affecting text must be in command mode (press Esc twice). Here are some of the commands can be used to move around one character at a time

Commands and their description

•**k:** Moves the cursor up one line

•**j:** Moves the cursor down one line

•**h:** Moves the cursor to the left one character position

•**I:** Moves the cursor to the right one character position

•**0 or \:** Positions cursor at beginning of live

•**$:** Positions cursor at end of line

Practical-7

Aim - Write a program do add and subtract too numbers using shell Programming

File Syntax:- $ vi addsub sh

Program :-

```
echo "addition and subtraction in shell Programming="
echo "Enter two numbers" :-
read num1
read num2
sum = 'expr $num1 + $num2'
diff = 'expr $ num1 - $ num2"
echo " Sum of $numl and $num2 is $ sum"
echo "Difference of $num1 and frum2 is $ diff"
```

```
“Addition and Subtraction in shell Programming:-“
“Enter two numbers:-“
67
54
“Sum of 67 and 54 is 121”
“Difference of 67 and 54 is 13”
```

Practical-8

Aim :- Write a program do multiply two numbers using Shell Programming

File Syntax: - $ vi multiply.sh

Program:-

```
echo" Multiplication in shell Programming:-"
echo Enter First Number: "
read num1
echo Enter Second Number:"
read num2
resell = 'expr $num1 = $num2"
echo Product of $num1 and $num2 is $result"
```

```
“Multiplication in shell programming :-”
“Enter First Number:-”
6
“Enter Second Number:-“
7
“Product of 6 and 7 is 42”
```

Practical-9

Aim:- Write a program to divide two numbers using shell Programming

File Syntax:- $ vi divide-sh

Program:-

```
echo" Division Program in shell Programming :-"
echo" Enter First Number:
read num1
echo "Enter Second Number?"
read num 2
result = 'expr $num1 / $num 2'
echo "Division of $ num1 and $num2 is $result"
```

```
"Division Program in shell
programming :-"
"Enter First Number:-"
45
"Enter second Number:-"
5
"Division of 45 and 5 is 9"
```

Practical-10

Aim:- Write a program to find out factorial of a number.

Syntax:- $ vi factorial-sh

Program :-

```
echo" Enter a number:"
read num
fact=1
while [$num-gt1]
do
        fact = $((fact * num)) # fact = fact num
        num = $((num-1)) #num = num-1
done
echo "factorial of $num is $ fact"
```

```
“Enter a number :-”
5
“Factorial of 5 is 120”
```